

Jay D. Suthar

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Mechanical engineering student with hands-on lab and shop experience: equipment setup and operation, hand and power tool use, and careful data collection. Looking to support hydraulic model construction, testing, and day-to-day lab operations.

EDUCATION

Colorado State University
Bachelor of Science: Mechanical Engineering, Aerospace Concentration

Expected Graduation – May 2027

INDUSTRY EXPERIENCE

Undergraduate Research Assistant, Walter Scott Jr. College of Engineering 01/2025 – 08/2025

- Supported experimental studies on laser-matter interactions in Dr. Jorge Rocca's lab through the SURE program
- Set up and maintained test equipment, coordinated sensor selection, material sourcing, and test workflows
- Collected and reduced experimental data, generated plots in MATLAB to support engineering decisions
- Presented findings to the research team and handled general lab upkeep and inventory tracking.

Undergraduate Teaching Assistant, Walter Scott Jr. College of Engineering 08/2024 – 05/2025

- Taught Excel and MATLAB workflows for MECH 103 and 105, supporting 60+ students
- Graded technical assignments, ran weekly office hours, and helped students debug analysis problems.

OURA Lab Researcher, Office of Undergraduate Research and Artistry 09/2024 – 10/2024

- Designed and ran a study on 3D-printing filament biodegradability and tensile strength, including sample procurement and prep
- Operated tensile testing equipment, recorded load and displacement data, and reduced results in Excel and MATLAB
- Maintained lab equipment between runs, managed sample inventory, and reported findings to a faculty mentor.

L'SPACE NPWEE Proposal Writer, National Aeronautic Space Administration (NASA) 05/2024 – 08/2024

- Collaborated with a 15+ member team and subject matter experts to develop and propose innovative solutions for NASA systems constraints, investing 200+ hours in research, meetings, and evaluations.

PROJECTS AND ORGANIZATIONS

Payload Bay Lead, Ram Launch Initiative, Colorado State University 08/2024 – Present

- Designed and analyzed the payload bay assembly using SolidWorks and Fusion 360 Electronics while integrating sensors (BMP280, PCF8523) and microcontrollers (Teensy 4.1, BNO055) for system functionality
- Contributed to the Preliminary Design Review (PDR) by conducting system analyses and optimizing avionics integration for aerospace applications
- Secured 7th place overall out of 68 teams participating in the competition.

President, Ram Rocketry Club, Colorado State University 08/2024 - Present

- Led mechanical design and fabrication of rockets, coordinating with machinists at CSU RPL
- Directed assembly and flight testing, applying engineering change processes (ECOs)
- Secured corporate funding to expand instrumentation and material resources for active stabilization projects.

Project Intern, SUPROS 07/2022 – 08/2022

- Assisted with prototype satellite development and mechanical fixture evaluations
- Researched prior designs and implemented feasible improvements under technical constraints.

SKILLS AND CERTIFICATIONS

Hand and power tool operation, physical model fabrication (composites, filament winding), lab equipment operation and maintenance, tensile testing, sample prep, data reduction (Excel, MATLAB, Python)

Tools and Certifications: SolidWorks, Fusion 360, Siemens NX, Autodesk Inventor, Arduino, basic FEA. SolidWorks CAD Design Associate (CSWA), HAAS CNC Basic Mill Operator Certificate, NASA L'SPACE NPWEE, College of Engineering Dean's List (FA23, SP24).